1.37 Integration: Definite Integrals and Areas

Worked Solutions

1a)

$$\int_2^6 (8x+1)\,dx = [4x^2+x]_2^6$$

$$= (4\times 6^2+6)-(4\times 2^2+2) = (150)-(18) = 132$$

1b)

$$\int_3^4 (6x^2-1)\,dx = [2x^3-x]_3^4$$

$$= (2\times 4^3-4)-(2\times 3^3-3) = (124)-(51) = 73$$

1c)

$$\int_4^8 (500+4x-x^3)\,dx = \left[500x+2x^2-\frac{1}{4}x^4\right]_4^8$$

$$= \left(500\times 8+2\times 8^2-\frac{1}{4}\times 8^4\right)-\left(500\times 4+2\times 4^2-\frac{1}{4}\times 4^4\right)$$

$$= (3104)-(1968) = 1136$$

1d)

$$\int_{-1}^1 (14x-9x^5)\,dx = \left[7x^2-\frac{3}{2}x^6\right]_{-1}^1$$

$$= \left(7\times 1^2-\frac{3}{2}\times 1^6\right)-\left(7\times (-1)^2-\frac{3}{2}\times (-1)^6\right)$$

$$= \left(\frac{11}{2}\right)-\left(\frac{11}{2}\right) = 0$$

2)

$$\int_1^2 (10x-x^4)\,dx = \left[5x^2-\frac{1}{5}x^5\right]_1^2$$

$$= \left(5\times 2^2-\frac{1}{5}\times 2^5\right)-\left(5\times 1^2-\frac{1}{5}\times 1^5\right)$$

$$= \left(\frac{68}{5}\right)-\left(\frac{24}{5}\right) = \frac{44}{5}$$

3)

$$\int_{-2}^{2}(-2-x^2+x^3-x^4)\,dx=\left[-2x-\frac{1}{3}x^3+\frac{1}{4}x^4-\frac{1}{5}x^5\right]_{-2}^{2}$$
$$=\left(-2\times 2-\frac{1}{3}\times 2^3+\frac{1}{4}\times 2^4-\frac{1}{5}\times 2^5\right)$$
$$-\left(-2\times(-2)-\frac{1}{3}\times(-2)^3+\frac{1}{4}\times(-2)^4-\frac{1}{5}\times(-2)^5\right)$$
$$=\left(-\frac{136}{15}\right)-\left(\frac{256}{15}\right)=-\frac{392}{15}$$
$$\therefore \text{Area}=\frac{392}{15}$$

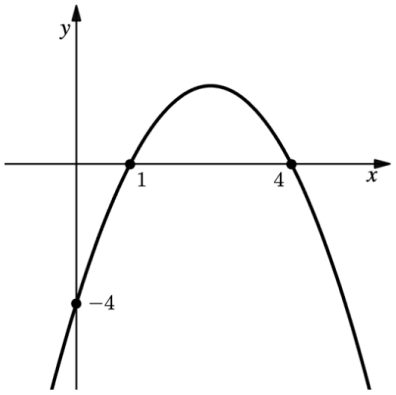
4)

$$\int_1^3(f(x)+2)\,dx=50\Rightarrow\int_1^3 f(x)\,dx+\int_1^3 2\,dx=50$$
$$\Rightarrow\int_1^3 f(x)\,dx+[2x]_1^3=50$$
$$\Rightarrow\int_1^3 f(x)\,dx+(2\times 3)-(2\times 1)=50$$
$$\Rightarrow\int_1^3 f(x)\,dx+4=50\Rightarrow\int_1^3 f(x)\,dx+4=46$$

5)

$$\int_{-1/2}^{1/2}(10-x^5+4x^3)\,dx=\left[10x-\frac{1}{6}x^6+x^4\right]_{-1/2}^{1/2}$$
$$=\left(10\times\frac{1}{2}-\frac{1}{6}\times\left(\frac{1}{2}\right)^6+\left(\frac{1}{2}\right)^4\right)$$
$$-\left(10\times\left(-\frac{1}{2}\right)-\frac{1}{6}\times\left(-\frac{1}{2}\right)^6+\left(-\frac{1}{2}\right)^4\right)$$
$$=\left(\frac{1943}{384}\right)-\left(-\frac{1897}{384}\right)=10$$

6a)



6b)

Part of the area is below the x-axis. The integral of a region below the x-axis is negative, so $\int_1^5 (x-1)(4-x)\,dx$ is the sum of a positive part and a negative part, and not the sum of two positive values, which is want we want for the area.

7)

$$\int_1^k (4x - 2x^3)\,dx = 0 \Rightarrow \left[2x^2 - \frac{1}{2}x^4\right]_1^k = 0$$

$$\Rightarrow \left(2 \times k^2 - \frac{1}{2} \times k^4\right) - \left(2 \times 1^2 - \frac{1}{2} \times 1^4\right) = 0$$

$$\Rightarrow \left(2k^2 - \frac{1}{2}k^4\right) - \left(\frac{3}{2}\right) = 0 \Rightarrow -\frac{1}{2}k^4 + 2k^2 - \frac{3}{2} = 0$$

$$\Rightarrow k^4 - 4k^2 + 3 = 0 \Rightarrow (k^2 - 3)(k^2 - 1) = 0$$

$$\Rightarrow k = \pm\sqrt{3} \text{ or } k = \pm 1$$

$$\therefore k = \sqrt{3} \text{ as } k > 1$$

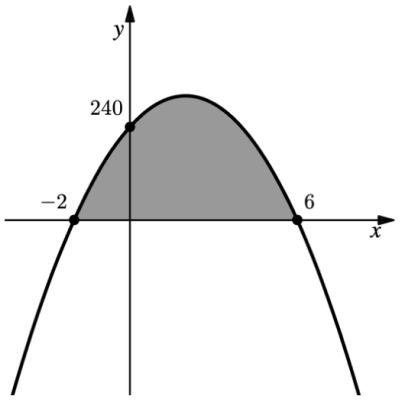
8)

$$y = k(x+2)(x-6)$$

When $x = 0$, $y = 240$:

$$240 = k(0+2)(0-6) \Rightarrow k = -20$$

$$y = -20(x+2)(x-6) \Rightarrow y = -20x^2 + 80x + 240$$



$$\int_{-2}^{6} (-20x^2 + 80x + 240)\,dx = \left[-\frac{20}{3}x^3 + 40x^2 + 240x\right]_{-2}^{6}$$

$$= \left(-\frac{20}{3} \times 6^3 + 40 \times 6^2 + 240 \times 6\right)$$

$$-\left(-\frac{20}{3} \times (-2)^3 + 40 \times (-2)^2 + 240 \times (-2)\right)$$

$$= (1440) - \left(-\frac{800}{3}\right) = \frac{5120}{3}$$

9)

$$\int_{-3}^{3}(kx^2-5x^4)\,dx=18\Rightarrow\left[\frac{1}{3}kx^3-x^5\right]_{-3}^{3}=18$$

$$\Rightarrow\left(\frac{1}{3}k\times3^3-3^5\right)-\left(\frac{1}{3}k\times(-3)^3-(-3)^5\right)=18$$

$$\Rightarrow(9k-243)-(-9k+243)=18$$

$$\Rightarrow18k-486=18\Rightarrow k=28$$

10)

$$\int_{-2}^{0}x^5\,dx=\left[\frac{1}{6}x^6\right]_{-2}^{0}=\left(\frac{1}{6}\times0^6\right)-\left(\frac{1}{6}\times(-2)^6\right)=-\frac{32}{3}$$

$$\int_{0}^{2}x^5\,dx=\left[\frac{1}{6}x^6\right]_{0}^{2}=\left(\frac{1}{6}\times2^6\right)-\left(\frac{1}{6}\times0^6\right)=\frac{32}{3}$$

$$\therefore\text{Area}=\frac{32}{3}-\left(-\frac{32}{3}\right)=\frac{64}{3}$$

Alternatively (using symmetry):

$$\text{Area}=2\int_{0}^{2}x^5\,dx=2\left[\frac{1}{6}x^6\right]_{0}^{2}$$

$$=2\left(\left(\frac{1}{6}\times2^6\right)-\left(\frac{1}{6}\times0^6\right)\right)=2\left(\frac{32}{3}\right)=\frac{64}{3}$$

11)

$$4-x^2=0\Rightarrow x^2=4\Rightarrow x=\pm2$$

$$\int_{-4}^{-2}(4-x^2)\,dx=\left[4x-\frac{1}{3}x^3\right]_{-4}^{-2}$$

$$=\left(4\times(-2)-\frac{1}{3}\times(-2)^3\right)-\left(4\times(-4)-\frac{1}{3}\times(-4)^3\right)$$

$$=\left(-\frac{16}{3}\right)-\left(\frac{16}{3}\right)=-\frac{32}{3}$$

$$\int_{-2}^{2}(4-x^2)\,dx=\left[4x-\frac{1}{3}x^3\right]_{-2}^{2}$$

$$=\left(4\times2-\frac{1}{3}\times2^3\right)-\left(4\times(-2)-\frac{1}{3}\times(-2)^3\right)$$

$$=\left(\frac{16}{3}\right)-\left(-\frac{16}{3}\right)=\frac{32}{3}$$

$$\int_{2}^{4}(4-x^2)\,dx=\left[4x-\frac{1}{3}x^3\right]_{2}^{4}$$

$$=\left(4\times4-\frac{1}{3}\times4^3\right)-\left(4\times2-\frac{1}{3}\times2^3\right)$$

$$=\left(-\frac{16}{3}\right)-\left(\frac{16}{3}\right)=-\frac{32}{3}$$

$$\therefore \text{Area} = \frac{32}{3} - \left(-\frac{32}{3}\right) - \left(-\frac{32}{3}\right) = 32$$

Alternative solution using symmetry:

$$\int_{-4}^{-2} (4 - x^2)\, dx = \left[4x - \frac{1}{3}x^3\right]_{-4}^{-2}$$

$$= \left(4 \times (-2) - \frac{1}{3} \times (-2)^3\right) - \left(4 \times (-4) - \frac{1}{3} \times (-4)^3\right)$$

$$= \left(-\frac{16}{3}\right) - \left(\frac{16}{3}\right) = -\frac{32}{3}$$

$$\int_{-2}^{0} (4 - x^2)\, dx = \left[4x - \frac{1}{3}x^3\right]_{-2}^{0}$$

$$= \left(4 \times 0 - \frac{1}{3} \times 0^3\right) - \left(4 \times (-2) - \frac{1}{3} \times (-2)^3\right)$$

$$= (0) - \left(-\frac{16}{3}\right) = \frac{16}{3}$$

$$\text{Area} = 2\left(\frac{16}{3} - \left(-\frac{32}{3}\right)\right) = 32$$

12)

$$\int_{\sqrt{2}}^{\sqrt{3}} (3x^2 + 1)\, dx = [x^3 + x]_{\sqrt{2}}^{\sqrt{3}}$$

$$= \left((\sqrt{3})^3 + \sqrt{3}\right) - \left((\sqrt{2})^3 + \sqrt{2}\right)$$

$$= (3\sqrt{3} + \sqrt{3}) - (2\sqrt{2} + \sqrt{2}) = 4\sqrt{3} - 3\sqrt{2}$$

13)

$$\int_{1}^{1+\sqrt{2}} (50 - 6x^2)\, dx = [50x - 2x^3]_{1}^{1+\sqrt{2}}$$

$$= \left(50(1 + \sqrt{2}) - 2(1 + \sqrt{2})^3\right) - (50 \times 1 - 2 \times 1^3)$$

$$= \left(50 + 50\sqrt{2} - 2(1 + 3\sqrt{2} + 6 + 2\sqrt{2})\right) - (48)$$

$$= 2 + 50\sqrt{2} - 2(7 + 5\sqrt{2}) = 2 + 50\sqrt{2} - 14 - 10\sqrt{2} = 40\sqrt{2} - 12$$

14)

$$\int_{1+\sqrt{3}}^{2+\sqrt{3}} (3x^2 + 6x)\, dx = [x^3 + 3x^2]_{1+\sqrt{3}}^{2+\sqrt{3}}$$

$$= \left((2 + \sqrt{3})^3 + 3(2 + \sqrt{3})^2\right) - \left((1 + \sqrt{3})^3 + 3(1 + \sqrt{3})^2\right)$$

$$= \left((8 + 12\sqrt{3} + 18 + 3\sqrt{3}) + 3(4 + 4\sqrt{3} + 3)\right)$$

$$- \left((1 + 3\sqrt{3} + 9 + 3\sqrt{3}) + 3(1 + 2\sqrt{3} + 3)\right)$$

$$= (47 + 27\sqrt{3}) - (22 + 12\sqrt{3}) = 25 + 15\sqrt{3}$$

15a)

$$\int_4^{16} \sqrt{x}\,dx = \int_4^{16} x^{1/2}\,dx = \left[\frac{2}{3}x^{3/2}\right]_4^{16}$$
$$= \left(\frac{2}{3}\times 16^{3/2}\right) - \left(\frac{2}{3}\times 4^{3/2}\right) = \left(\frac{128}{3}\right) - \left(\frac{16}{3}\right) = \frac{112}{3}$$

15b)

$$\int_{-2}^{-1} \frac{6}{x^2}\,dx = \int_{-2}^{-1} 6x^{-2}\,dx = [-6x^{-1}]_{-2}^{-1} = \left[-\frac{6}{x}\right]_{-2}^{-1}$$
$$= \left(-\frac{6}{-1}\right) - \left(-\frac{6}{-2}\right) = (6) - (3) = 3$$

15c)

$$\int_1^9 \frac{4}{\sqrt{x}}\,dx = \int_1^9 4x^{-1/2}\,dx = [8x^{1/2}]_1^9$$
$$= (8\times 9^{1/2}) - (8\times 1^{1/2}) = (24) - (8) = 16$$

15d)

$$\int_1^{64} \left(\frac{1}{\sqrt[3]{x}} - \frac{1}{\sqrt[4]{x}}\right)dx = \int_1^{64} (x^{-1/3} - x^{-1/4})\,dx$$
$$= \left[\frac{3}{2}x^{2/3} - \frac{4}{3}x^{3/4}\right]_1^{64}$$
$$= \left(\frac{3}{2}\times 64^{2/3} - \frac{4}{3}\times 64^{3/4}\right) - \left(\frac{3}{2}\times 1^{2/3} - \frac{4}{3}\times 1^{3/4}\right)$$
$$= \left(24 - \frac{64}{3}\sqrt{2}\right) - \left(\frac{3}{2} - \frac{4}{3}\right) = \frac{143}{6} - \frac{64}{3}\sqrt{2}$$

16)

$$\int_0^1 \sqrt[4]{x}\,dx = \int_0^1 x^{1/4}\,dx = \left[\frac{4}{5}x^{5/4}\right]_0^1$$
$$= \left(\frac{4}{5}\times 1^{5/4}\right) - \left(\frac{4}{5}\times 0^{5/4}\right) = \left(\frac{4}{5}\right) - (0) = \frac{4}{5}$$

17)

$$\int_1^8 \frac{12}{\sqrt[3]{x}}\,dx = \int_1^8 12x^{-1/3}\,dx = [18x^{2/3}]_1^8$$
$$= (18\times 8^{2/3}) - (18\times 1^{2/3}) = (72) - (18) = 54$$

18)

$$\int_{-5}^{-3} \frac{9}{x^2}\,dx = \int_{-5}^{-3} 9x^{-2}\,dx = [-9x^{-1}]_{-5}^{-3}$$

$$= (-9 \times (-3)^{-1}) - (-9 \times (-5)^{-1}) = (3) - \left(\frac{9}{5}\right) = \frac{6}{5}$$

19)

$$\int_{-3}^{-2} (x^3 + 4x^2 + x - 6)\,dx = \left[\frac{1}{4}x^4 + \frac{4}{3}x^3 + \frac{1}{2}x^2 - 6x\right]_{-3}^{-2}$$

$$= \left(\frac{1}{4} \times (-2)^4 + \frac{4}{3} \times (-2)^3 + \frac{1}{2} \times (-2)^2 - 6 \times (-2)\right)$$

$$- \left(\frac{1}{4} \times (-3)^4 + \frac{4}{3} \times (-3)^3 + \frac{1}{2} \times (-3)^2 - 6 \times (-3)\right)$$

$$= \left(\frac{22}{3}\right) - \left(\frac{27}{4}\right) = \frac{7}{12}$$

$$\int_{-2}^{1} (x^3 + 4x^2 + x - 6)\,dx = \left[\frac{1}{4}x^4 + \frac{4}{3}x^3 + \frac{1}{2}x^2 - 6x\right]_{-2}^{1}$$

$$= \left(\frac{1}{4} \times 1^4 + \frac{4}{3} \times 1^3 + \frac{1}{2} \times 1^2 - 6 \times 1\right)$$

$$- \left(\frac{1}{4} \times (-2)^4 + \frac{4}{3} \times (-2)^3 + \frac{1}{2} \times (-2)^2 - 6 \times (-2)\right)$$

$$= \left(-\frac{47}{12}\right) - \left(\frac{22}{3}\right) = -\frac{45}{4}$$

Shaded area $= \frac{7}{12} + \frac{45}{4} = \frac{71}{6}$.

20)

$$\int_{k}^{2k} \left(\frac{1}{2}x^2 + 3\right)\,dx = \left[\frac{1}{6}x^3 + 3x\right]_{k}^{2k}$$

$$= \left(\frac{1}{6} \times (2k)^3 + 3 \times (2k)\right) - \left(\frac{1}{6} \times k^3 + 3 \times k\right)$$

$$= \left(\frac{4}{3}k^3 + 6k\right) - \left(\frac{1}{6}k^3 + 3k\right) = \frac{7}{6}k^3 + 3k$$

21)

$$x(5 - x) = x \Rightarrow 5x - x^2 = x \Rightarrow 0 = x^2 - 4x \Rightarrow 0 = x(x - 4) \Rightarrow x = 0 \text{ or } x = 4$$

$$\int_0^4 x(5 - x)\,dx = \int_0^4 (5x - x^2)\,dx = \left[\frac{5}{2}x^2 - \frac{1}{3}x^3\right]_0^4$$

$$= \left(\frac{5}{2} \times 4^2 - \frac{1}{3} \times 4^3\right) - \left(\frac{5}{2} \times 0^2 - \frac{1}{3} \times 0^3\right) = \left(\frac{56}{3}\right) - (0) = \frac{56}{3}$$

When $x = 4$, $y = 4$.

Area of triangle $= \frac{1}{2} \times 4 \times 4 = 8$.

$$\therefore \text{Area} = \frac{56}{3} - 8 = \frac{32}{3}$$

Alternatively:

$$\int_0^4 x(5-x)\,dx - \int_0^4 x\,dx$$

$$= \int_0^4 (5x - x^2)\,dx - \int_0^4 x\,dx = \int_0^4 (5x - x^2 - x)\,dx$$

$$= \int_0^4 (4x - x^2)\,dx = \left[2x^2 - \frac{1}{3}x^3\right]_0^4$$

$$= \left(2 \times 4^2 - \frac{1}{3} \times 4^3\right) - \left(2 \times 0^2 - \frac{1}{3} \times 0^3\right) = \left(\frac{32}{3}\right) - (0) = \frac{32}{3}$$

22)

$$20 - x^2(x-3) = 20 - 4x \Rightarrow -x^2(x-3) = -4x$$

$$\Rightarrow x^2(x-3) = 4x \Rightarrow x^3 - 3x^2 - 4x = 0$$

$$\Rightarrow x(x^2 - 3x - 4) = 0 \Rightarrow x(x-4)(x+1) = 0$$

$$\Rightarrow x = 0 \text{ or } x = 4 \text{ or } x = -1$$

$$\int_0^4 (20 - x^2(x-3))\,dx = \int_0^4 (20 - x^3 + 3x^2)\,dx = \left[20x - \frac{1}{4}x^4 + x^3\right]_0^4$$

$$= \left(20 \times 4 - \frac{1}{4} \times 4^4 + 4^3\right) - \left(20 \times 0 - \frac{1}{4} \times 0^4 + 0^3\right)$$

$$= (80) - (0) = 80$$

When $x = 0$, $y = 20 - 4 \times 0 = 20$.

When $x = 4$, $y = 20 - 4 \times 4 = 4$.

Area of trapezium $= \frac{1}{2} \times 4 \times (20 + 4) = 48$.

$$\therefore \text{Area} = 80 - 48 = 32$$

Alternatively:

$$\int_0^4 (20 - x^2(x-3))\,dx - \int_0^4 (20 - 4x)\,dx$$

$$= \int_0^4 (20 - x^3 + 3x^2)\,dx - \int_0^4 (20 - 4x)\,dx$$

$$= \int_0^4 (20 - x^3 + 3x^2 - (20 - 4x))\,dx = \int_0^4 (-x^3 + 3x^2 + 4x)\,dx$$

$$= \left[-\frac{1}{4}x^4 + x^3 + 2x^2\right]_0^4$$

$$= \left(-\frac{1}{4} \times 4^4 + 4^3 + 2 \times 4^2\right) - \left(-\frac{1}{4} \times 0^4 + 0^3 + 2 \times 0^2\right)$$

$$= (32) - (0) = 32$$

23)

$$\int_1^2 \left(\frac{k}{x^3}+2\right)\,dx = 5 \Rightarrow \int_1^2 (kx^{-3}+2)\,dx = 5$$
$$\Rightarrow \left[\frac{kx^{-2}}{-2}+2x\right]_1^2 = 5$$
$$\Rightarrow \left(-\frac{1}{8}k+4\right)-\left(-\frac{k}{2}+2\right) = 5$$
$$\Rightarrow \frac{3}{8}k+2 = 5 \Rightarrow k = 8$$

24)

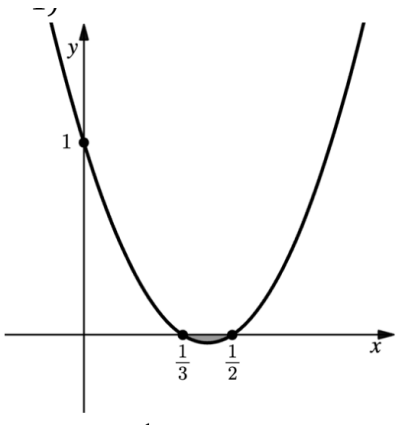
$$\int_1^9 (2kx-3\sqrt{x})\,dx = 108 \Rightarrow \int_1^9 (2kx-3x^{1/2})\,dx = 108$$
$$\Rightarrow [kx^2-2x^{3/2}]_1^9 = 108$$
$$\Rightarrow (k\times 9^2-2\times 9^{3/2})-(k\times 1^2-2\times 1^{3/2}) = 108$$
$$\Rightarrow (81k-54)-(k-2) = 108 \Rightarrow 80k-52 = 108 \Rightarrow k = 2$$

25)

$$\int_1^{16} (k\sqrt{x}+2x+2)\,dx = 3k^2 \Rightarrow \int_1^{16} (kx^{1/2}+2x+2)\,dx = 3k^2$$
$$\Rightarrow \left[\frac{2}{3}kx^{3/2}+x^2+2x\right]_1^{16} = 3k^2$$
$$\Rightarrow \left(\frac{2}{3}k\times 16^{3/2}+16^2+2\times 16\right)-\left(\frac{2}{3}k\times 1^{3/2}+1^2+2\times 1\right) = 3k^2$$
$$\Rightarrow \left(\frac{128}{3}k+288\right)-\left(\frac{2}{3}k+3\right) = 3k^2$$
$$\Rightarrow 42k+285 = 3k^2 \Rightarrow 3k^2-42k-285 = 0$$
$$\Rightarrow k^2-14k-95 = 0 \Rightarrow (k+5)(k-19) = 0 \Rightarrow k = -5 \text{ or } k = 19$$

26)

$$6x^2-5x+1 = (3x-1)(2x-1)$$



$$\int_{1/3}^{1/2}(6x^2-5x+1)\,dx=\left[2x^3-\frac{5}{2}x^2+x\right]_{1/3}^{1/2}$$
$$=\left(2\times\left(\frac{1}{2}\right)^3-\frac{5}{2}\times\left(\frac{1}{2}\right)^2+\frac{1}{2}\right)$$
$$-\left(2\times\left(\frac{1}{3}\right)^3-\frac{5}{2}\times\left(\frac{1}{3}\right)^2+\frac{1}{3}\right)$$
$$=\left(\frac{1}{8}\right)-\left(\frac{7}{54}\right)=-\frac{1}{216}$$
$$\therefore \text{Area}=\frac{1}{216}$$

27)

$$\int_2^4(3p-2p^2x)\,dx=[3px-p^2x^2]_2^4$$
$$=(3p\times4-p^2\times4^2)-(3p\times2-p^2\times2^2)$$
$$=(12p-16p^2)-(6p-4p^2)=6p-12p^2$$
$$=-12\left(p^2-\frac{1}{2}p\right)=-12\left(\left(p-\frac{1}{4}\right)^2-\frac{1}{16}\right)$$
$$=-12\left(p-\frac{1}{4}\right)^2+\frac{3}{4}$$

So the maximum occurs when $p=\frac{1}{4}$, and the maximum is $\frac{3}{4}$.

28)

$$\int_4^k\left(\frac{5}{x^3}-\frac{1}{x^2}\right)dx=0\Rightarrow\int_4^k(5x^{-3}-x^{-2})\,dx=0$$
$$\Rightarrow\left[\frac{5x^{-2}}{-2}-\frac{x^{-1}}{-1}\right]_4^k=0\Rightarrow\left[-\frac{5}{2}x^{-2}+x^{-1}\right]_4^k=0$$
$$\Rightarrow\left(-\frac{5}{2}k^{-2}+k^{-1}\right)-\left(-\frac{5}{2}\times4^{-2}+4^{-1}\right)=0$$
$$\Rightarrow\left(\frac{1}{k}-\frac{5}{2k^2}\right)-\left(\frac{3}{32}\right)=0\Rightarrow k-\frac{5}{2}-\frac{3k^2}{32}=0$$
$$\Rightarrow 32k-80-3k^2=0\Rightarrow 3k^2-32k+80=0$$
$$\Rightarrow(3k-20)(k-4)=0\Rightarrow k=\frac{20}{3}\text{ or }k=4$$
$$\therefore k=\frac{20}{3}\text{ as }k>4$$

29)

$$\int_1^k \left(\frac{30}{\sqrt{x}} + 10\right) dx = 6090 \Rightarrow \int_1^k (30x^{-1/2} + 10)\, dx = 6090$$

$$\Rightarrow [60x^{1/2} + 10x]_1^k = 6090$$

$$\Rightarrow (60k^{1/2} + 10k) - (60 \times 1^{1/2} + 10 \times 1) = 6090$$

$$\Rightarrow (60k^{1/2} + 10k) - (70) = 6090$$

$$\Rightarrow 10k + 60k^{1/2} - 6160 = 0 \Rightarrow k + 6k^{1/2} - 616 = 0$$

$$\Rightarrow (k^{1/2} + 28)(k^{1/2} - 22) = 0 \Rightarrow k^{1/2} = 22 \Rightarrow k = 484$$

30)

$$x\sqrt{x} - 8 = 0 \Rightarrow x^{3/2} - 8 = 0 \Rightarrow x^{3/2} = 8 \Rightarrow x = 8^{2/3} = 4$$

$$\int_0^4 (x\sqrt{x} - 8)\, dx = \int_0^4 (x^{3/2} - 8)\, dx = \left[\frac{2}{5}x^{5/2} - 8x\right]_0^4$$

$$= \left(\frac{2}{5} \times 4^{5/2} - 8 \times 4\right) - \left(\frac{2}{5} \times 0^{5/2} - 8 \times 0\right) = \left(-\frac{96}{5}\right) - (0) = -\frac{96}{5}$$

$$\int_4^9 (x\sqrt{x} - 8)\, dx = \int_4^9 (x^{3/2} - 8)\, dx = \left[\frac{2}{5}x^{5/2} - 8x\right]_4^9$$

$$= \left(\frac{2}{5} \times 9^{5/2} - 8 \times 9\right) - \left(\frac{2}{5} \times 4^{5/2} - 8 \times 4\right) = \left(\frac{126}{5}\right) - \left(-\frac{96}{5}\right) = \frac{222}{5}$$

$$\therefore \text{Area} = \frac{222}{5} - \left(-\frac{96}{5}\right) = \frac{318}{5}$$

31)

$$\int_{-2}^{-1} \left(x + \frac{1}{x^2}\right) dx = \int_{-2}^{-1} (x + x^{-2})\, dx$$

$$= \left[\frac{1}{2}x^2 + \frac{x^{-1}}{-1}\right]_{-2}^{-1} = \left[\frac{1}{2}x^2 - \frac{1}{x}\right]_{-2}^{-1}$$

$$= \left(\frac{1}{2} \times (-1)^2 - \frac{1}{-1}\right) - \left(\frac{1}{2} \times (-2)^2 - \frac{1}{-2}\right) = \left(\frac{3}{2}\right) - \left(\frac{5}{2}\right) = -1$$

$$\int_1^2 \left(x + \frac{1}{x^2}\right) dx = \int_1^2 (x + x^{-2})\, dx$$

$$= \left[\frac{1}{2}x^2 + \frac{x^{-1}}{-1}\right]_1^2 = \left[\frac{1}{2}x^2 - \frac{1}{x}\right]_1^2$$

$$= \left(\frac{1}{2} \times 2^2 - \frac{1}{2}\right) - \left(\frac{1}{2} \times 1^2 - \frac{1}{1}\right) = \left(\frac{3}{2}\right) - \left(-\frac{1}{2}\right) = 2$$

$$\therefore \text{Area} = 2 - (-1) = 3$$

32)

$$\int_1^2 \left(\frac{1}{x^2} - \frac{6}{x^3} + \frac{11}{x^4} - \frac{6}{x^5}\right) dx = \int_1^2 (x^{-2} - 6x^{-3} + 11x^{-4} - 6x^{-5}) dx$$

$$= \left[\frac{x^{-1}}{-1} - \frac{6x^{-2}}{-2} + \frac{11x^{-3}}{-3} - \frac{6x^{-4}}{-4}\right]_1^2 = \left[-\frac{1}{x} + \frac{3}{x^2} - \frac{11}{3x^3} + \frac{3}{2x^4}\right]_1^2$$

$$= \left(-\frac{1}{2} + \frac{3}{2^2} - \frac{11}{3 \times 2^3} + \frac{3}{2 \times 2^4}\right) - \left(-\frac{1}{1} + \frac{3}{1^2} - \frac{11}{3 \times 1^3} + \frac{3}{2 \times 1^4}\right)$$

$$= \left(-\frac{11}{96}\right) - \left(-\frac{1}{6}\right) = \frac{5}{96}$$

$$\int_2^3 \left(\frac{1}{x^2} - \frac{6}{x^3} + \frac{11}{x^4} - \frac{6}{x^5}\right) dx = \int_2^3 (x^{-2} - 6x^{-3} + 11x^{-4} - 6x^{-5}) dx$$

$$= \left[\frac{x^{-1}}{-1} - \frac{6x^{-2}}{-2} + \frac{11x^{-3}}{-3} - \frac{6x^{-4}}{-4}\right]_2^3 = \left[-\frac{1}{x} + \frac{3}{x^2} - \frac{11}{3x^3} + \frac{3}{2x^4}\right]_2^3$$

$$= \left(-\frac{1}{3} + \frac{3}{3^2} - \frac{11}{3 \times 3^3} + \frac{3}{2 \times 3^4}\right) - \left(-\frac{1}{2} + \frac{3}{2^2} - \frac{11}{3 \times 2^3} + \frac{3}{2 \times 2^4}\right)$$

$$= \left(-\frac{19}{162}\right) - \left(-\frac{11}{96}\right) = -\frac{7}{2592}$$

$$\therefore \text{Area} = \frac{5}{96} - \left(-\frac{7}{2592}\right) = \frac{71}{1296}$$

33)

$$3k - \int_0^3 35(x-3)(x-5)\sqrt{x}\, dx = 300\sqrt{3}$$

$$\Rightarrow 3k - \int_0^3 35(x^2 - 8x + 15)\sqrt{x}\, dx = 300\sqrt{3}$$

$$\Rightarrow 3k - \int_0^3 (35x^{5/2} - 280x^{3/2} + 525x^{1/2}) dx = 300\sqrt{3}$$

$$\Rightarrow 3k - \left[10x^{7/2} - 112x^{5/2} + 350x^{3/2}\right]_0^3 = 300\sqrt{3}$$

$$\Rightarrow 3k - \left(\left(10 \times 3^{7/2} - 112 \times 3^{5/2} + 350 \times 3^{3/2}\right) - \left(10 \times 0^{7/2} - 112 \times 0^{5/2} + 350 \times 0^{3/2}\right)\right) = 300\sqrt{3}$$

$$\Rightarrow 3k - (270\sqrt{3} - 1008\sqrt{3} + 1050\sqrt{3}) = 300\sqrt{3}$$

$$\Rightarrow 3k - 312\sqrt{3} = 300\sqrt{3}$$

$$\Rightarrow 3k = 612\sqrt{3} \Rightarrow k = 204\sqrt{3}$$

34)

$$\int_k^{9k} \left(\frac{2}{\sqrt{x}} + x\right) dx = \int_k^{9k} (2x^{-1/2} + x)$$

$$= \left[4x^{1/2} + \frac{1}{2}x^2\right]_k^{9k}$$

$$= \left(4 \times (9k)^{1/2} + \frac{1}{2} \times (9k)^2\right) - \left(4k^{1/2} + \frac{1}{2}k^2\right)$$

$$= \left(12k^{1/2} + \frac{81}{2}k^2\right) - \left(4k^{1/2} + \frac{1}{2}k^2\right) = 8k^{1/2} + 40k^2$$

35)

$$\frac{2}{x^2} = 3 - x \Rightarrow 2 = 3x^2 - x^3 \Rightarrow x^3 - 3x^2 + 2 = 0$$

$$\Rightarrow (x-1)(x^2 - 2x - 2) = 0 \Rightarrow x = 1 \text{ or } x = 1 \pm \sqrt{3}$$

$$\int_1^{1+\sqrt{3}} \frac{2}{x^2}\,dx = \int_1^{1+\sqrt{3}} 2x^{-2}dx = \left[\frac{2x^{-1}}{-1}\right]_1^{1+\sqrt{3}}$$

$$= \left[-\frac{2}{x}\right]_1^{1+\sqrt{3}} = \left(-\frac{2}{1+\sqrt{3}}\right) - \left(-\frac{2}{1}\right)$$

$$= (1 - \sqrt{3}) - (-2) = 3 - \sqrt{3}$$

When $x = 1$, $y = 3 - 1 = 2$.
When $x = 1 + \sqrt{3}$, $y = 3 - (1 + \sqrt{3}) = 2 - \sqrt{3}$.

$$\text{Area of trapezium} = \frac{1}{2} \times \sqrt{3}(2 + 2 - \sqrt{3}) = \frac{-3 + 4\sqrt{3}}{2}$$

$$\therefore \text{Area} = \frac{-3 + 4\sqrt{3}}{2} - (3 - \sqrt{3}) = \frac{-9 + 6\sqrt{3}}{2}$$

36(a))

$$\int_1^3 \frac{x+2}{x^3}\,dx = \int_1^3 \left(\frac{x}{x^3} + \frac{2}{x^3}\right)dx = \int_1^3 (x^{-2} + 2x^{-3})dx$$

$$= \left[\frac{x^{-1}}{-1} + \frac{2x^{-2}}{-2}\right]_1^3 = \left[-\frac{1}{x} - \frac{1}{x^2}\right]_1^3$$

$$= \left(-\frac{1}{3} - \frac{1}{3^2}\right) - \left(-\frac{1}{1} - \frac{1}{1^2}\right) = \left(-\frac{4}{9}\right) - (-2) = \frac{14}{9}$$

36(b))

$$\int_2^4 \frac{3x+8}{x^4}\,dx = \int_2^4 \left(\frac{3x}{x^4} + \frac{8}{x^4}\right)dx = \left[3x^{-3} + 8x^{-4}\right]_2^4$$

$$= \left[\frac{3x^{-2}}{-2} + \frac{8x^{-3}}{-3}\right]_2^4 = \left[-\frac{3}{2x^2} - \frac{8}{3x^3}\right]_2^4$$

$$= \left(-\frac{3}{2 \times 4^2} - \frac{8}{3 \times 4^3}\right) - \left(-\frac{3}{2 \times 2^2} - \frac{8}{3 \times 2^3}\right)$$

$$= \left(-\frac{13}{96}\right) - \left(-\frac{17}{24}\right) = \frac{55}{96}$$

36(c))

$$\int_{-3}^{-1} \frac{4-5x}{2x^3}\,dx = \int_{-3}^{-1} \left(\frac{4}{2x^3} - \frac{5x}{2x^3}\right) dx$$

$$= \int_{-3}^{-1} \left(2x^{-3} - \frac{5}{2}x^{-2}\right) dx = \left[\frac{2x^{-2}}{-2} - \frac{5}{2} \times \frac{x^{-1}}{-1}\right]_{-3}^{-1} = \left[-\frac{1}{x^2} + \frac{5}{2x}\right]_{-3}^{-1}$$

$$= \left(-\frac{1}{(-1)^2} + \frac{5}{2\times(-1)}\right) - \left(-\frac{1}{(-3)^2} + \frac{5}{2\times(-3)}\right)$$

$$= \left(-\frac{7}{2}\right) - \left(-\frac{17}{18}\right) = -\frac{23}{9}$$

36(d))

$$\int_{-4}^{-2} \frac{3x^2-2}{6x^4}\,dx = \int_{-4}^{-2} \left(\frac{3x^2}{6x^4} - \frac{2}{6x^4}\right) dx$$

$$= \int_{-4}^{-2} \left(\frac{1}{2}x^{-2} - \frac{1}{3}x^{-4}\right) dx = \left[\frac{1}{2} \times \frac{x^{-1}}{-1} - \frac{1}{3} \times \frac{x^{-3}}{-3}\right]_{-4}^{-2} = \left[-\frac{1}{2x} + \frac{1}{9x^3}\right]_{-4}^{-2}$$

$$= \left(-\frac{1}{2\times(-2)} + \frac{1}{9\times(-2)^3}\right) - \left(-\frac{1}{2\times(-4)} + \frac{1}{9\times(-4)^3}\right)$$

$$= \left(\frac{17}{72}\right) - \left(\frac{71}{576}\right) = \frac{65}{576}$$

37)

$$\int_2^3 \frac{x^2+10}{x^2}\,dx = \int_2^3 \left(\frac{x^2}{x^2} + \frac{10}{x^2}\right) dx = \int_2^3 (1+10x^{-2})dx$$

$$= \left[x + \frac{10x^{-1}}{-1}\right]_2^3 = \left[x - \frac{10}{x}\right]_2^3 = \left(3 - \frac{10}{3}\right) - \left(2 - \frac{10}{2}\right)$$

$$= \left(-\frac{1}{3}\right) - (-3) = \frac{8}{3}$$

38)

$$\int_4^9 \frac{\sqrt{x}-2}{\sqrt{x}}\,dx = \int_4^9 \left(\frac{\sqrt{x}}{\sqrt{x}} - \frac{2}{\sqrt{x}}\right) dx = \int_4^9 (1-2x^{-1/2})dx$$

$$= \left[x - \frac{2x^{1/2}}{1/2}\right]_4^9 = \left[x - 4x^{1/2}\right]_4^9$$

$$= (9 - 4\times 9^{1/2}) - (4 - 4\times 4^{1/2}) = (-3) - (-4) = 1$$

39)

$$\int_{-5}^{-3} \frac{x+3}{x^3}\,dx = \int_{-5}^{-3} \left(\frac{x}{x^3} + \frac{3}{x^3}\right) dx$$

$$= \int_{-5}^{-3} (x^{-2} + 3x^{-3})dx = \left[\frac{x^{-1}}{-1} + \frac{3x^{-2}}{-2}\right]_{-5}^{-3} = \left[-\frac{1}{x} - \frac{3}{2x^2}\right]_{-5}^{-3}$$

$$= \left(-\frac{1}{-3} - \frac{3}{2\times(-3)^2}\right) - \left(-\frac{1}{-5} - \frac{3}{2\times(-5)^2}\right)$$

$$= \left(\frac{1}{6}\right) - \left(\frac{7}{50}\right) = \frac{2}{75}$$

40)

$$\int_1^2 \frac{kx+1}{x^4}\,dx = \frac{11}{12} \Rightarrow \int_1^2 \left(\frac{kx}{x^4} + \frac{1}{x^4}\right) dx = \frac{11}{12}$$

$$\Rightarrow \int_1^2 (kx^{-3} + x^{-4})dx = \frac{11}{12} \Rightarrow \left[\frac{kx^{-2}}{-2} + \frac{x^{-3}}{-3}\right]_1^2 = \frac{11}{12}$$

$$\Rightarrow \left[-\frac{k}{2x^2} - \frac{1}{3x^3}\right]_1^2 = \frac{11}{12}$$

$$\Rightarrow \left(-\frac{k}{2\times 2^2} - \frac{1}{3\times 2^3}\right) - \left(-\frac{k}{2\times 1^2} - \frac{1}{3\times 1^3}\right) = \frac{11}{12}$$

$$\Rightarrow \left(-\frac{1}{8}k - \frac{1}{24}\right) - \left(-\frac{1}{2}k - \frac{1}{3}\right) = \frac{11}{12}$$

$$\Rightarrow \frac{3}{8}k + \frac{7}{24} = \frac{11}{12} \Rightarrow k = \frac{5}{3}$$

41)

$$\int_4^9 \frac{4-k\sqrt{x}}{6\sqrt{x}}\,dx = 1 \Rightarrow \int_4^9 \left(\frac{4}{6\sqrt{x}} - \frac{k\sqrt{x}}{6\sqrt{x}}\right) dx = 1$$

$$\Rightarrow \int_4^9 \left(\frac{2}{3}x^{-1/2} - \frac{1}{6}k\right) dx = 1 \Rightarrow \left[\frac{\frac{2}{3}x^{1/2}}{1/2} - \frac{1}{6}kx\right]_4^9 = 1$$

$$\Rightarrow \left[\frac{4}{3}x^{1/2} - \frac{1}{6}kx\right]_4^9 = 1$$

$$\Rightarrow \left(\frac{4}{3}\times 9^{1/2} - \frac{1}{6}k\times 9\right) - \left(\frac{4}{3}\times 4^{1/2} - \frac{1}{6}k\times 4\right) = 1$$

$$\Rightarrow \left(4 - \frac{3}{2}k\right) - \left(\frac{8}{3} - \frac{2}{3}k\right) = 1 \Rightarrow \frac{4}{3} - \frac{5}{6}k = 1 \Rightarrow k = \frac{2}{5}$$

42)

$$\int_{1/2}^{k} \frac{x^2-1}{x^2}\,dx = 0 \Rightarrow \int_{1/2}^{k} \left(\frac{x^2}{x^2} - \frac{1}{x^2}\right) dx = 0$$

$$\Rightarrow \int_{1/2}^{k} (1 - x^{-2}) dx = 0 \Rightarrow \left[x - \frac{x^{-1}}{-1}\right]_{1/2}^{k} = 0$$

$$\Rightarrow \left[x + \frac{1}{x}\right]_{1/2}^{k} = 0 \Rightarrow \left(k + \frac{1}{k}\right) - \left(\frac{1}{2} + 2\right) = 0$$

$$\Rightarrow k + \frac{1}{k} - \frac{5}{2} = 0 \Rightarrow 2k^2 + 2 - 5k = 0$$

$$\Rightarrow (2k-1)(k-2) = 0 \Rightarrow k = \frac{1}{2} \text{ or } k = 2$$

$$\therefore k = 2 \text{ as } k > \frac{1}{2}$$

43)

$$\frac{(3-x)(5-x)}{x^4} = 0 \Rightarrow x = 3 \text{ or } x = 5$$

$$\int_{5/2}^{3} \frac{(3-x)(5-x)}{x^4}\,dx = \int_{5/2}^{3} \frac{x^2 - 8x + 15}{x^4}\,dx$$

$$= \int_{5/2}^{3} \left(\frac{x^2}{x^4} - \frac{8x}{x^4} + \frac{15}{x^4}\right) dx = \int_{5/2}^{3} (x^{-2} - 8x^{-3} + 15x^{-4}) dx$$

$$= \left[\frac{x^{-1}}{-1} - \frac{8x^{-2}}{-2} + \frac{15x^{-3}}{-3}\right]_{5/2}^{3} = \left[-\frac{1}{x} + \frac{4}{x^2} - \frac{5}{x^3}\right]_{5/2}^{3}$$

$$= \left(-\frac{1}{3} + \frac{4}{3^2} - \frac{5}{3^3}\right) - \left(-\frac{1}{5/2} + \frac{4}{(5/2)^2} - \frac{5}{(5/2)^3}\right)$$

$$= \left(-\frac{2}{27}\right) - \left(-\frac{2}{25}\right) = \frac{4}{675}$$

$$\int_{3}^{5} \frac{(3-x)(5-x)}{x^4}\,dx = \left[-\frac{1}{x} + \frac{4}{x^2} - \frac{5}{x^3}\right]_{3}^{5}$$

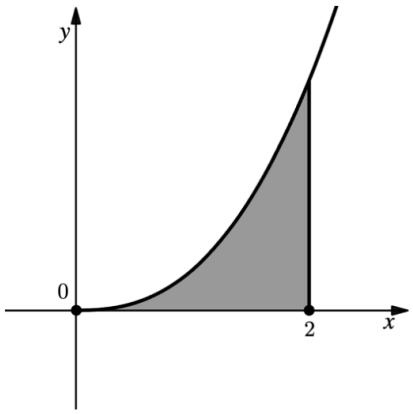
$$= \left(-\frac{1}{5} + \frac{4}{5^2} - \frac{5}{5^3}\right) - \left(-\frac{1}{3} + \frac{4}{3^2} - \frac{5}{3^3}\right) = \left(-\frac{2}{25}\right) - \left(-\frac{2}{27}\right) = -\frac{4}{675}$$

$$\int_{5}^{6} \frac{(3-x)(5-x)}{x^4}\,dx = \left[-\frac{1}{x} + \frac{4}{x^2} - \frac{5}{x^3}\right]_{5}^{6}$$

$$= \left(-\frac{1}{6} + \frac{4}{6^2} - \frac{5}{6^3}\right) - \left(-\frac{1}{5} + \frac{4}{5^2} - \frac{5}{5^3}\right) = \left(-\frac{17}{216}\right) - \left(-\frac{2}{25}\right) = \frac{7}{5400}$$

$$\therefore \text{Area} = \frac{4}{675} - \left(-\frac{4}{675}\right) + \frac{7}{5400} = \frac{71}{5400}$$

44)



$$\int_0^2 84x^{5/2}dx = \left[\frac{84x^{7/2}}{7/2}\right]_0^2 = \left[24x^{7/2}\right]_0^2$$

$$= (24 \times 2^{7/2}) - (24 \times 0^{7/2}) = (192\sqrt{2}) - (0) = 192\sqrt{2}$$

45)

$$\int_k^{4k} \frac{3x-1}{x\sqrt{x}}\,dx = \int_k^{4k} \frac{3x-1}{x^{3/2}}\,dx = \int_k^{4k} \left(\frac{3x}{x^{3/2}} - \frac{1}{x^{3/2}}\right)dx$$

$$= \int_k^{4k} (3x^{-1/2} - x^{-3/2})dx = \left[\frac{3x^{1/2}}{1/2} - \frac{x^{-1/2}}{-1/2}\right]_k^{4k} = \left[6x^{1/2} + 2x^{-1/2}\right]_k^{4k}$$

$$= \left(6 \times (4k)^{1/2} + 2 \times (4k)^{-1/2}\right) - \left(6k^{1/2} + 2k^{-1/2}\right)$$

$$= \left(12k^{1/2} + k^{-1/2}\right) - \left(6k^{1/2} + 2k^{-1/2}\right)$$

$$= 6k^{1/2} - k^{-1/2} = \frac{6k^{1/2}}{1} - \frac{1}{k^{1/2}} = \frac{6k-1}{k^{1/2}} = \frac{6k-1}{\sqrt{k}} \text{ as required}$$

46)

$$3x + 14y = 47 \Rightarrow y = \frac{47}{14} - \frac{3}{14}x$$

$$\frac{47}{14} - \frac{3}{14}x = \frac{x+1}{\sqrt{x}} \Rightarrow 47 - 3x = \frac{14x+14}{\sqrt{x}} \Rightarrow 47\sqrt{x} - 3x\sqrt{x} = 14x + 14 \Rightarrow 47x^{1/2} - 3x^{3/2} =$$

$$14x + 14 \Rightarrow 3x^{3/2} + 14x - 47x^{1/2} + 14 = 0 \Rightarrow 3(x^{1/2})^3 + 14(x^{1/2})^2 - 47x^{1/2} + 14 = 0 \Rightarrow$$

$$(3x^{1/2} - 1)(x^{1/2} + 7)(x^{1/2} - 2) = 0 \Rightarrow x^{1/2} = \frac{1}{3} \text{ or } x^{1/2} = -7 \text{ or } x^{1/2} = 2$$

So $x = \frac{1}{9}$ or $x = 4$ as $x^{1/2} \neq -7$.

$$\int_{1/9}^4 \frac{x+1}{\sqrt{x}}\,dx = \int_{1/9}^4 \left(\frac{x}{\sqrt{x}} + \frac{1}{\sqrt{x}}\right)dx = \int_{1/9}^4 (x^{1/2} + x^{-1/2})dx$$

$$= \left[\frac{x^{3/2}}{3/2} + \frac{x^{1/2}}{1/2}\right]_{1/9}^4 = \left[\frac{2}{3}x^{3/2} + 2x^{1/2}\right]_{1/9}^4$$

$$= \left(\frac{2}{3} \times 4^{3/2} + 2 \times 4^{1/2}\right) - \left(\frac{2}{3} \times (\frac{1}{9})^{3/2} + 2 \times (\frac{1}{9})^{1/2}\right) = \left(\frac{28}{3}\right) - \left(\frac{56}{81}\right) = \frac{700}{81}$$

When $x = \frac{1}{9}$, $y = \frac{10}{3}$.

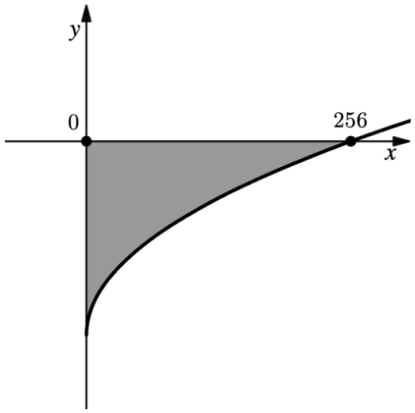
When $x = 4$, $y = \frac{5}{2}$.

$$\text{Area of Trapezium} = \frac{1}{2} \times \left(4 - \frac{1}{9}\right) \times \left(\frac{10}{3} + \frac{5}{2}\right) = \frac{1225}{108}$$

$$\therefore \text{Area} = \frac{1225}{108} - \frac{700}{81} = \frac{875}{324}$$

47)

$$0 = \sqrt{x} - 16 \Rightarrow \sqrt{x} = 16 \Rightarrow x = 256$$



$$\int_0^{256} \frac{x - 16\sqrt{x}}{\sqrt{x}}\, dx = \int_0^{256} (\sqrt{x} - 16) dx$$

$$= \int_0^{256} (x^{1/2} - 16) dx = \left[\frac{x^{3/2}}{3/2} - 16x\right]_0^{256} = \left[\frac{2}{3}x^{3/2} - 16x\right]_0^{256}$$

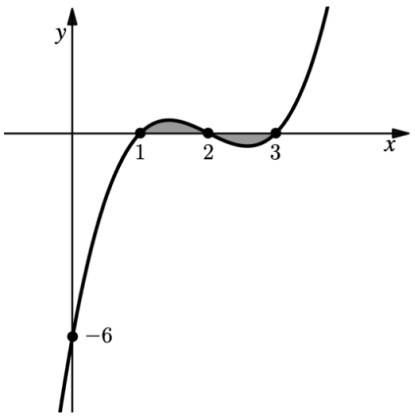
$$= \left(\frac{2}{3} \times 256^{3/2} - 16 \times 256\right) - \left(\frac{2}{3} \times 0^{3/2} - 16 \times 0\right)$$

$$= \left(-\frac{4096}{3}\right) - (0) = -\frac{4096}{3}$$

$$\therefore \text{Area} = \frac{4096}{3}$$

48(a))

$$(x-1)(x-2)(x-3) \equiv (x^2-3x+2)(x-3) \equiv x^3-6x^2+11x-6$$



$$\int_1^2 (x^3-6x^2+11x-6)dx = \left[\frac{1}{4}x^4-2x^3+\frac{11}{2}x^2-6x\right]_1^2$$

$$= \left(\frac{1}{4}\times 2^4-2\times 2^3+\frac{11}{2}\times 2^2-6\times 2\right)-\left(\frac{1}{4}\times 1^4-2\times 1^3+\frac{11}{2}\times 1^2-6\times 1\right)$$

$$= (-2)-\left(-\frac{9}{4}\right) = \frac{1}{4}$$

$$\int_2^3 (x^3-6x^2+11x-6)dx = \left[\frac{1}{4}x^4-2x^3+\frac{11}{2}x^2-6x\right]_2^3$$

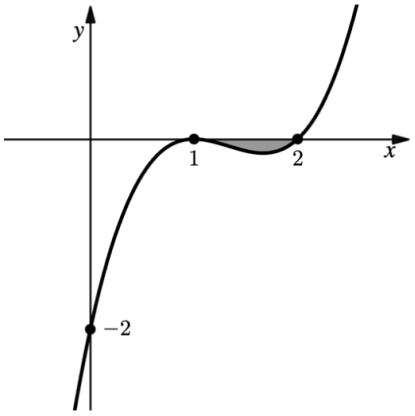
$$= \left(\frac{1}{4}\times 3^4-2\times 3^3+\frac{11}{2}\times 3^2-6\times 3\right)-\left(\frac{1}{4}\times 2^4-2\times 2^3+\frac{11}{2}\times 2^2-6\times 2\right)$$

$$= \left(-\frac{9}{4}\right)-(-2) = -\frac{1}{4}$$

$$\therefore \text{Area} = \frac{1}{4}+\frac{1}{4} = \frac{1}{2}$$

48(b))

$$(x-1)^2(x-2) \equiv (x^2-2x+1)(x-2) \equiv x^3-4x^2+5x-2$$



$$\int_1^2 (x^3 - 4x^2 + 5x - 2)dx = \left[\frac{1}{4}x^4 - \frac{4}{3}x^3 + \frac{5}{2}x^2 - 2x\right]_1^2$$

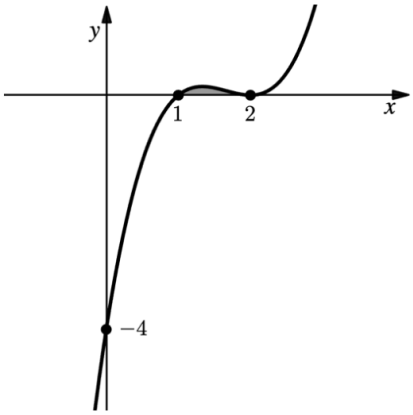
$$= \left(\frac{1}{4} \times 2^4 - \frac{4}{3} \times 2^3 + \frac{5}{2} \times 2^2 - 2 \times 2\right) - \left(\frac{1}{4} \times 1^4 - \frac{4}{3} \times 1^3 + \frac{5}{2} \times 1^2 - 2 \times 1\right)$$

$$= \left(-\frac{2}{3}\right) - \left(-\frac{7}{12}\right) = -\frac{1}{12}$$

$$\therefore \text{Area} = \frac{1}{12}$$

48(c))

$$(x-1)(x-2)^2 \equiv (x-1)(x^2 - 4x + 4) \equiv x^3 - 5x^2 + 8x - 4$$



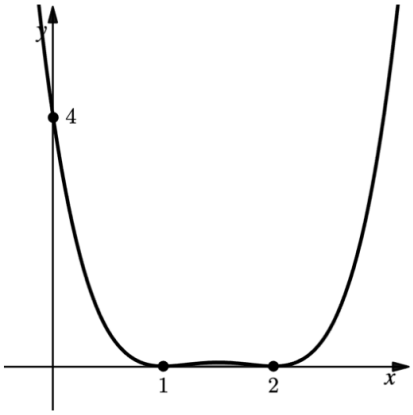
$$\int_1^2 (x^3 - 5x^2 + 8x - 4)dx = \left[\frac{1}{4}x^4 - \frac{5}{3}x^3 + 4x^2 - 4x\right]_1^2$$

$$= \left(\frac{1}{4} \times 2^4 - \frac{5}{3} \times 2^3 + 4 \times 2^2 - 4 \times 2\right) - \left(\frac{1}{4} \times 1^4 - \frac{5}{3} \times 1^3 + 4 \times 1^2 - 4 \times 1\right)$$

$$= \left(-\frac{4}{3}\right) - \left(-\frac{17}{12}\right) = \frac{1}{12}$$

48(d))

$$(x-1)^2(x-2)^2 \equiv (x^2 - 2x + 1)(x^2 - 4x + 4) \equiv x^4 - 6x^3 + 13x^2 - 12x + 4$$

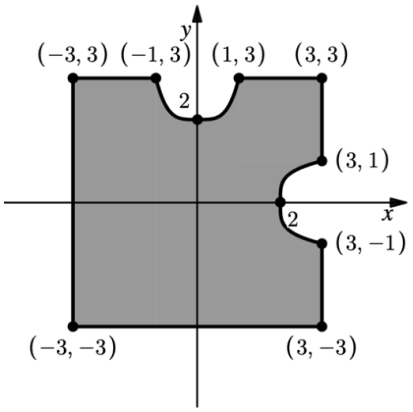


$$\int_1^2 (x^4 - 6x^3 + 13x^2 - 12x + 4)dx = \left[\frac{1}{5}x^5 - \frac{3}{2}x^4 + \frac{13}{3}x^3 - 6x^2 + 4x\right]_1^2$$

$$= \left(\frac{1}{5} \times 2^5 - \frac{3}{2} \times 2^4 + \frac{13}{3} \times 2^3 - 6 \times 2^2 + 4 \times 2\right) - \left(\frac{1}{5} \times 1^5 - \frac{3}{2} \times 1^4 + \frac{13}{3} \times 1^3 - 6 \times 1^2 + 4 \times 1\right)$$

$$= \left(\frac{16}{15}\right) - \left(\frac{31}{30}\right) = \frac{1}{30}$$

49)



$$2 \times 3 - \int_{-1}^{1} (x^4 + 2)dx = 6 - \left[\frac{1}{5}x^5 + 2x\right]_{-1}^{1}$$

$$= 6 - \left(\left(\frac{1}{5} \times 1^5 + 2 \times 1\right) - \left(\frac{1}{5} \times (-1)^5 + 2 \times (-1)\right)\right)$$

$$= 6 - \left(\left(\frac{11}{5}\right) - \left(-\frac{11}{5}\right)\right) = \frac{8}{5}$$

So the area of one of the gaps is $\frac{8}{5}$.

$$\therefore \text{Area} = 6 \times 6 - 2 \times \frac{8}{5} = 32.8 \text{ cm}^2$$

50)

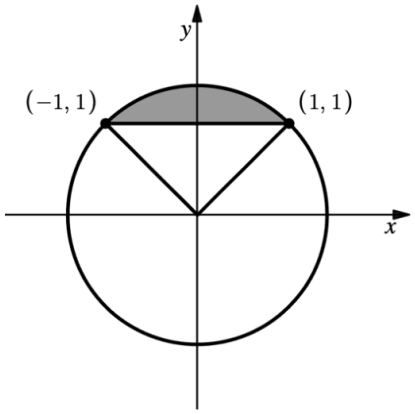
$$x^2 + (x^2)^2 = 2 \Rightarrow x^2 + x^4 = 2 \Rightarrow x^4 + x^2 - 2 = 0 \Rightarrow (x^2 + 2)(x^2 - 1) = 0 \Rightarrow x^2 = 1 \text{ as } x^2 \neq -2$$

So $x = 1$ or $x = -1$.

When $x = 1$, $y = 1$.

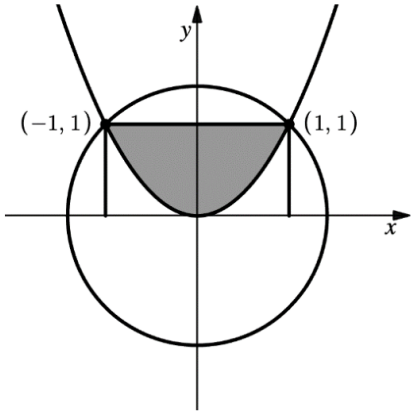
When $x = -1$, $y = 1$.

Focus on the segment first:



$$\text{Area of Segment} = \frac{1}{4} \times \pi \times (\sqrt{2})^2 - \frac{1}{2} \times \sqrt{2} \times \sqrt{2} = \frac{\pi}{2} - 1$$

Now focus on the integral:



$$2 \times 1 - \int_{-1}^{1} x^2 dx = 2 - \left[\frac{1}{3}x^3\right]_{-1}^{1}$$

$$= 2 - \left(\left(\frac{1}{3} \times 1^3\right) - \left(\frac{1}{3} \times (-1)^3\right)\right) = 2 - \left(\left(\frac{1}{3}\right) - \left(-\frac{1}{3}\right)\right) = \frac{4}{3}$$

So the total exact area is $(\frac{\pi}{2} - 1) + \left(\frac{4}{3}\right) = \frac{\pi}{2} + \frac{1}{3}$.

51)

$$y = 2\sqrt{x} \Rightarrow y = 2x^{1/2}$$

When $x = 4$, $y = 2 \times 4^{1/2} = 4$.

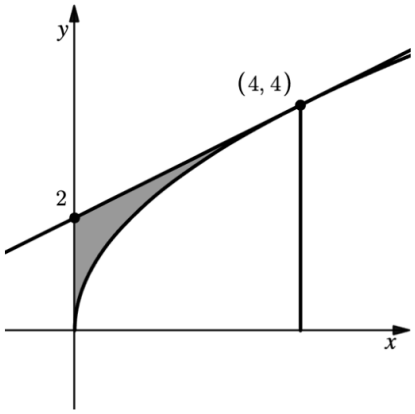
$$\frac{dy}{dx} = x^{-1/2} \qquad \left.\frac{dy}{dx}\right|_{x=4} = 4^{-1/2} = \frac{1}{2}$$

Tangent to the curve is $y - 4 = \frac{1}{2}(x - 4)$.

$$\Rightarrow y = \frac{1}{2}x + 2$$

$$\int_0^4 2x^{1/2} dx = \left[\frac{2x^{3/2}}{3/2}\right]_0^4 = \left[\frac{4}{3}x^{3/2}\right]_0^4 = \left(\frac{4}{3} \times 4^{3/2}\right) - \left(\frac{4}{3} \times 0^{3/2}\right) = \left(\frac{32}{3}\right) - (0) = \frac{32}{3}$$

$$\text{Area of Trapezium} = \frac{1}{2} \times 4 \times (2 + 4) = 12$$



$$\therefore \text{Area} = 12 - \frac{32}{3} = \frac{4}{3}$$

52)

$$y = (3 - x)(x + 2) \Rightarrow y = -x^2 + x + 6$$

When $x = 0$, $y = 6$ so $A(0, 6)$.

When $x = 2$, $y = 4$ so $B(2, 4)$.

$$\frac{dy}{dx} = -2x + 1 \qquad \left.\frac{dy}{dx}\right|_{x=0} = 1 \qquad \left.\frac{dy}{dx}\right|_{x=2} = -3$$

Normal at A: $y - 6 = -(x - 0) \Rightarrow y = -x + 6$.

Normal at B: $y - 4 = \frac{1}{3}(x - 2) \Rightarrow y = \frac{1}{3}x + \frac{10}{3}$.

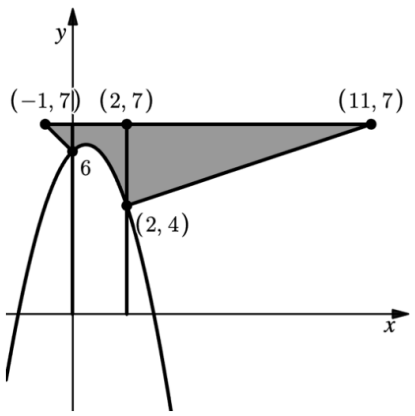
Normal at A intersects $y = 7$:

$$7 = -x + 6 \Rightarrow x = -1$$

Normal at B intersects $y = 7$:

$$7 = \frac{1}{3}x + \frac{10}{3} \Rightarrow x = 11$$

So endpoints of $y = 7$ are $(-1, 7)$ and $(11, 7)$.



$$1^{\text{st}} \text{ area} = \frac{1}{2} \times 1 \times 1 = \frac{1}{2}$$

$$2^{\text{nd}} \text{ area} = 2 \times 7 - \int_0^2 (-x^2 + x + 6)dx$$

$$= 14 - \left[-\frac{1}{3}x^3 + \frac{1}{2}x^2 + 6x\right]_0^2$$

$$= 14 - \left(\left(-\frac{1}{3} \times 2^3 + \frac{1}{2} \times 2^2 + 6 \times 2\right) - \left(-\frac{1}{3} \times 0^3 + \frac{1}{3} \times 0^2 + 6 \times 0\right)\right)$$

$$= 14 - \left(\left(\frac{34}{3}\right) - (0)\right) = \frac{8}{3}$$

$$3^{\text{rd}} \text{ area} = \frac{1}{2} \times 9 \times 3 = \frac{27}{2}$$

$$\therefore \text{Total area} = \frac{1}{2} + \frac{8}{3} + \frac{27}{2} = \frac{50}{3}$$

53)

$$y = 2x^3 + 15x^2 - 144x + 500 \qquad \frac{dy}{dx} = 6x^2 + 30x - 144$$

Turning points exist when $\frac{dy}{dx} = 0$:

$$6x^2 + 30x - 144 = 0 \Rightarrow x^2 + 5x - 24 = 0$$

$$\Rightarrow (x + 8)(x - 3) = 0 \Rightarrow x = -8 \text{ or } x = 3$$

$$\int_{-8}^{3} (2x^3 + 15x^2 - 144x + 500)dx = \left[\frac{1}{2}x^4 + 5x^3 - 72x^2 + 500x\right]_{-8}^{3}$$

$$= \left(\frac{1}{2} \times 3^4 + 5 \times 3^3 - 72 \times 3^2 + 500 \times 3\right) - \left(\frac{1}{2} \times (-8)^4 + 5 \times (-8)^3 - 72 \times (-8)^2 + 500 \times (-8)\right)$$

$$= \left(\frac{2055}{2}\right) - (-9120) = \frac{20295}{2} = 10147.5$$